

Zonghao(Johnny) Li

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Work Authorization Status: Green Card Holder (no sponsorships needed)

Education:

Boston University College of Engineering

Sept. 2022- May. 2026

Boston MA, Bachelor of Science, Mechanical Engineering

Experiences:

Voith Turbo

Shanghai, China

Engineering Center Intern.

June. 2025 - Sept. 2025

- Collaborated with cross-functional teams to develop and refine product documentation(user manuals)..
- Independently developed and launched a work hour tracking application used by **25 colleagues** in the Voith Shanghai Engineering Center Reducing Manual errors by **95%**.
- **Led the project** to develop an application utilizing **Power Apps** and **Power Automate** for sensors' Calibrations

DERS Entertainment

Brookline, MA

CAD Designer.

Oct. 2024 - May. 2025

- Designed and modeled 3D CAD components and assemblies using **SolidWorks** and **Blender**, preparing models for prototyping, rendering, and product presentation.
- Created and refined CAD geometry for visual clarity, manufacturability, and **rapid prototyping** through **3D printing**
- Managed multiple CAD design tasks while incorporating feedback into successive model iterations under tight production deadlines.

Projects:

Automated 2.5-DOF Linear Actuation Platform

- **Led a team of 4 People** to design and build an electromechanical Monopoly board using a 3 belt-drive **linear actuation system** modeled in **SolidWorks**, to enable 2.5 degrees of freedom of the system.
- **3D printed** components for **rapid prototyping** of different design ideas that were brought up by the team.
- Programmed motion sequences via **G-code**, achieving high accuracy and repeatable automated movement.

Optical Proximity Sensor Transduction System Design and Calibration

- Designed and calibrated an optical proximity sensor system to measure displacement response of a second-order spring-mass oscillator using **MATLAB/DAQ signal acquisition**.
- Analyzed dynamic system behavior through step-response and sinusoidal testing to determine natural frequency, damping ratio, spring constant, and sensor sensitivity.
- Performed uncertainty analysis and compared experimental frequency-response results with theoretical second-order oscillator models to validate system performance.

Multirole Fighter Jet Design

- Collaborated in a **four-member engineering team** to design a **multirole fighter** intended to replace the F-35 series.
- Defined and validated **military design requirements** per **MIL-STD-3013A** through Excel, targeting: **Max Mach 3.0, service ceiling 55,000 ft, radius 800 nm, and 25 deg/s instantaneous turn rate**.
- Created a full **3D parametric model in SolidWorks**, incorporating the **fuselage, blended-wing-body, twin-engine nacelles, and empennage**, and rendered using **SolidWorks Visualize**

Personal Commuter VTOL

- Led a 4-member engineering team in the conceptual and preliminary design of a **single rotor helicopter**
- Developed a detailed CAD model using SolidWorks and generated rendering using **SolidWorks Visualize**
- Built **Excel-based** parametric optimization models to evaluate design trade-offs, reducing projected **empty weight and structural volume**.

Skills:

Solidworks, Solidworks Visualize, Python, Matlab, Arduino, Xfoil, Onshape, Excel, Power Points, PowerApps, PowerAutomate, Soldering, Welding, Drill Press, 3D Printing, CNC Mill, Manual Mill, Manual Lathe, Laser Cutting, Sheet Metal Waterjet, Sand Casting, Basic Circuitry, Multimeter, Oscilloscope

Languages:

English, Chinese(Mandarin)

